Exam Date & Time: 29-Nov-2024 (09:30 AM - 12:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH END SEMESTER EXAMINATIONS, NOV/ DEC 2024
MEDICAL DEVICES (BME 2124)

NETWORK ANALYSIS [BME 2124]

Marks: 50 **Duration: 180 mins.**

Descriptive

Answer all the questions. Section Duration: 180 mins

1A) Determine the current I_o in the given circuit Figure Q1a

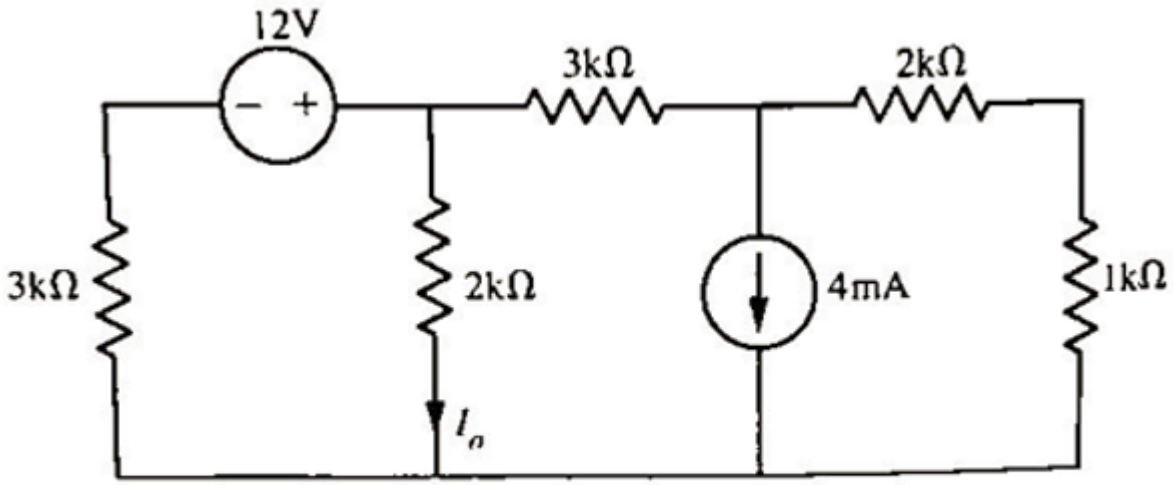


Figure Q1a

(4)

1B) Calculate the load R_L that will result in maximum power delivered to the load for the circuit of Figure Q1b. Also determine the maximum power. (3)

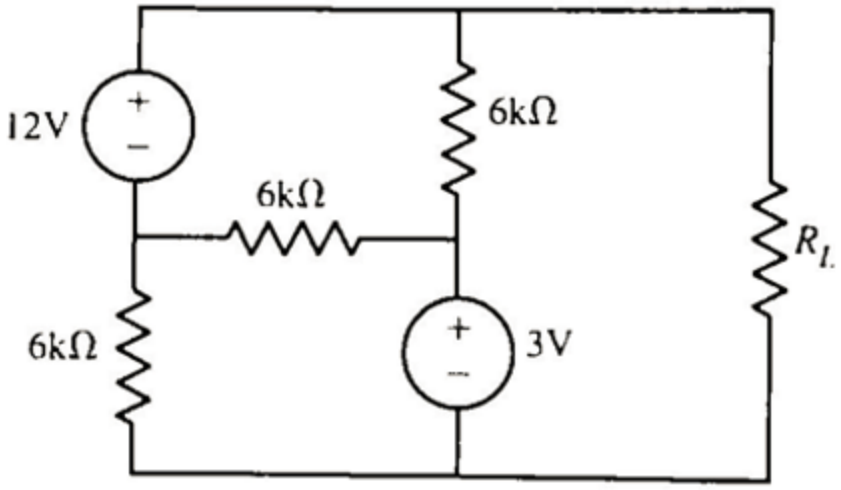
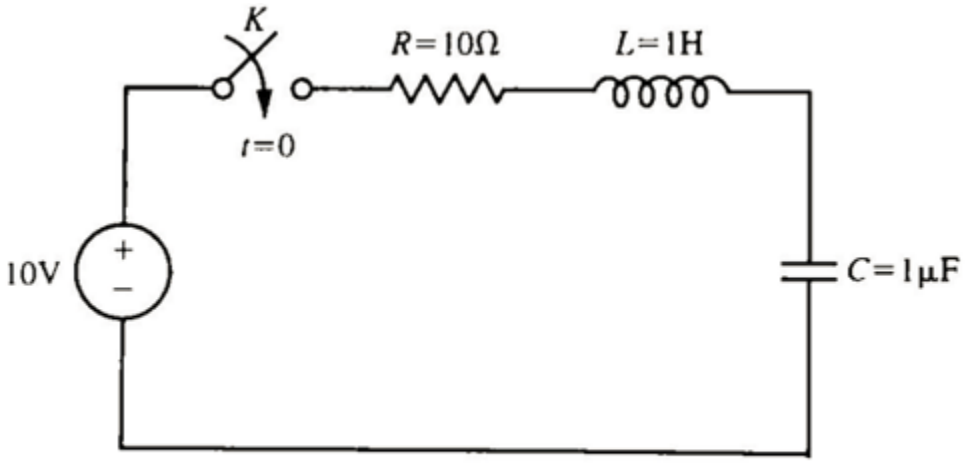


Figure Q1b

1C) In the network shown in Figure Q1c, the switch is closed at t=0, Determine i, $\frac{di}{dt}$ and $\frac{d^2i}{dt^2}$ at t = 0+



(3)

Figure Q1c

2A) In the network of Figure Q2a, the switch is moved from position 1 to position 2 at t=0. The steady-state having reached before switching. Calculate i, $\frac{di}{dt}$ and $\frac{d^2i}{dt^2}$ at t=0+. (5)

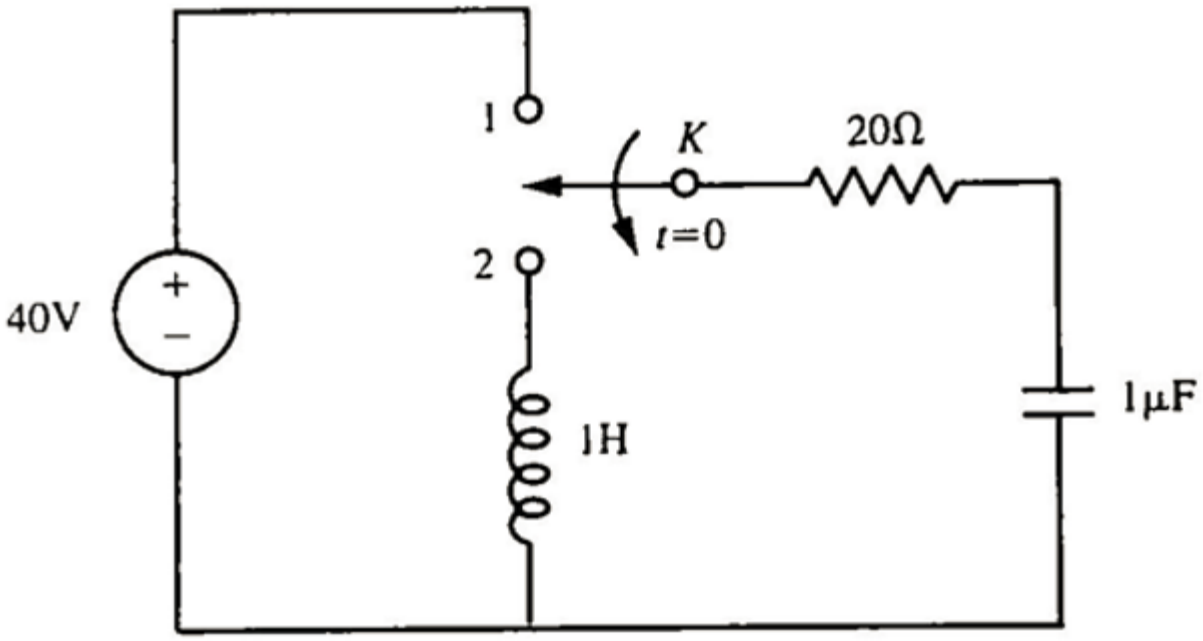


Figure Q2a

2B) A fresher in the Electronics circuits lab for the sake of curiosity sets up a series RLC network as shown in Figure Q2b. The capacitor can withstand very high voltages. Determine the voltage across the capacitor. Is it safe to touch the capacitor at resonance? Justify. (Hint: Quality factor $Q = \frac{\omega_0 L}{R}$) (3)

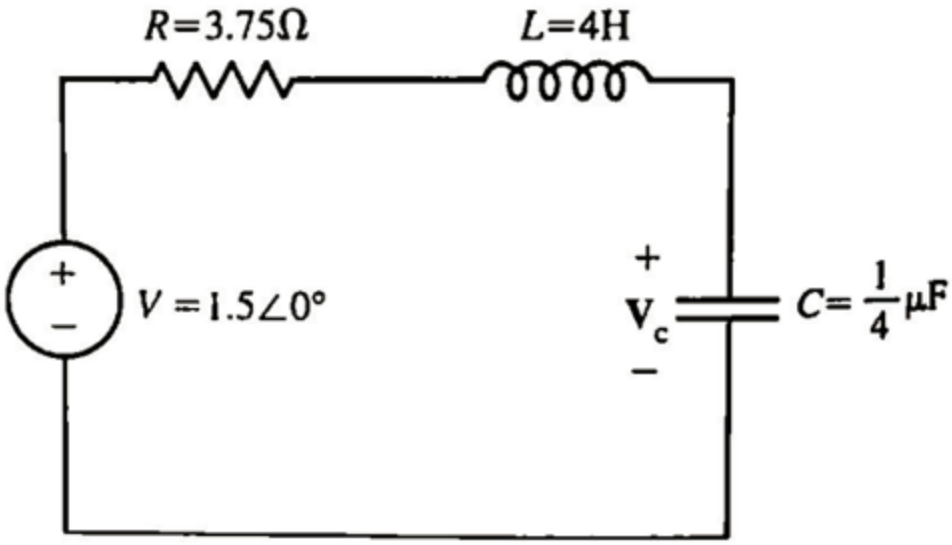
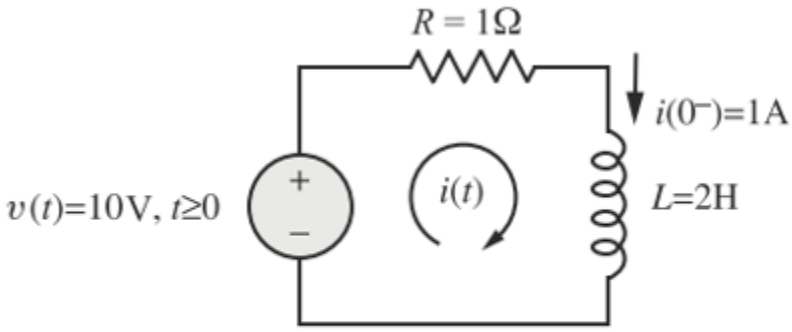


Figure Q2b

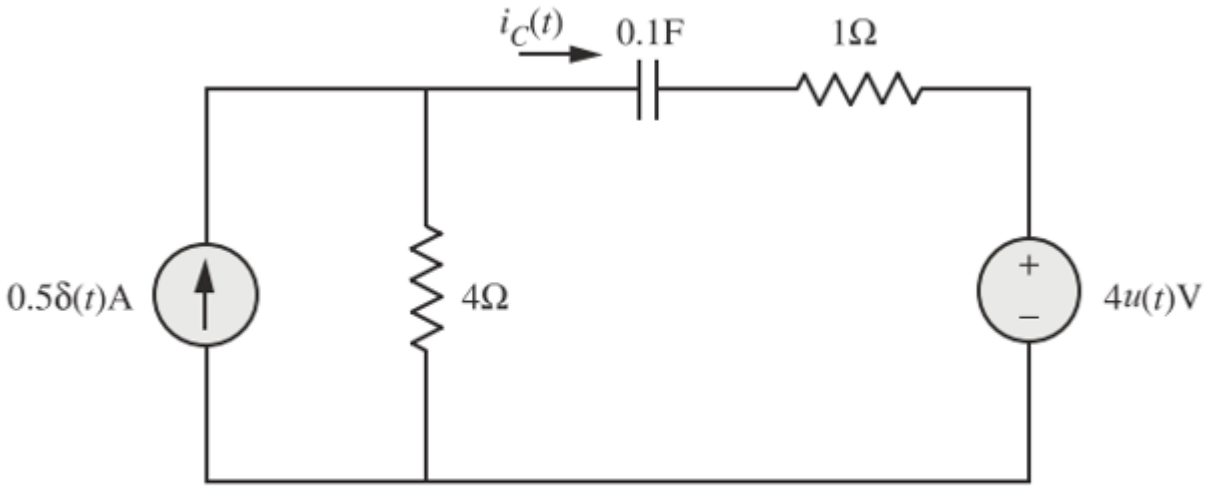
2C) Evaluate the current i(0) in the given network Figure Q2c using initial value theorem.



(2)

Figure Q2c

3A) Obtain a single integrodifferential equation in terms of i_C for the circuit of Figure Q3a. Solve for $I_C(s)$ using Laplace transform and then determine $i_c(t)$ by making use of inverse transform.



(4)

Figure Q3a

3B) i. Use properties of Laplace transform and solve for $x(t) = \sin(2\omega_0 t)\, u(t)$

ii. Evaluate the convolution of $h(t) = e^{-t}$ and $f(t) = e^{-2t}$ using Laplace transform (3)

3C) Calculate the inverse Laplace transform of

$X(s) = \frac{2s+4}{s^2+4s+3}$ (3)

4A) Analyze the two-port network depicted in Figure Q4a to determine its y-parameters. (4)

Then, using these parameters, calculate the current flowing through a 4Ω load connected to the output port when a 2A source is applied to the input port.

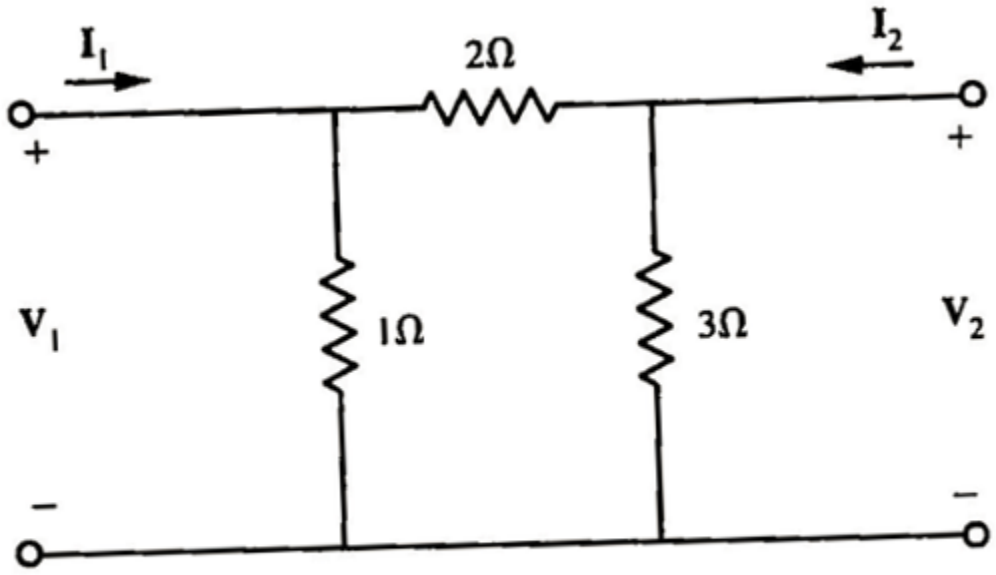


Figure Q4a

4B) Construct a circuit that realizes the following Z-parameters

$$z = \begin{bmatrix} 12 & 4 \\ 4 & 8 \end{bmatrix}$$ (3)

4C) Compute the hybrid parameters for the two-port network shown in Figure Q4c. (3)

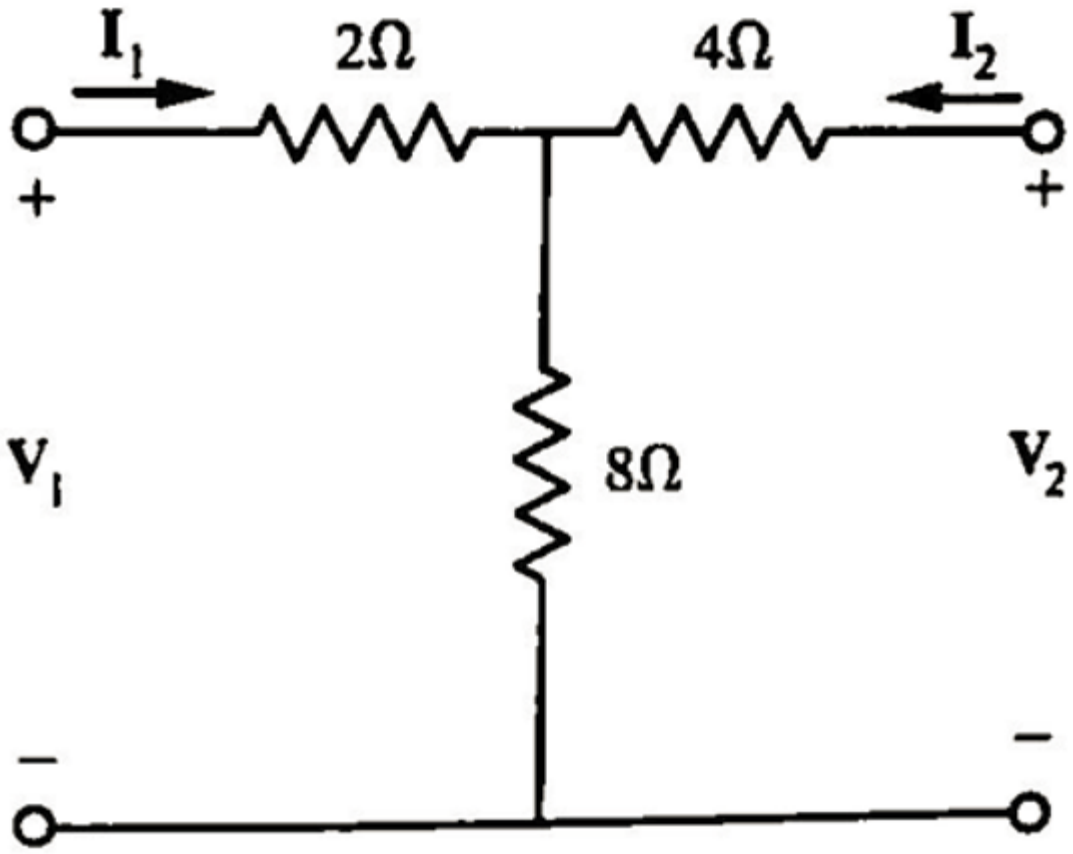


Figure Q4c

5A) Determine the Laplace transform of the waveform shown in Figure Q5a (4)

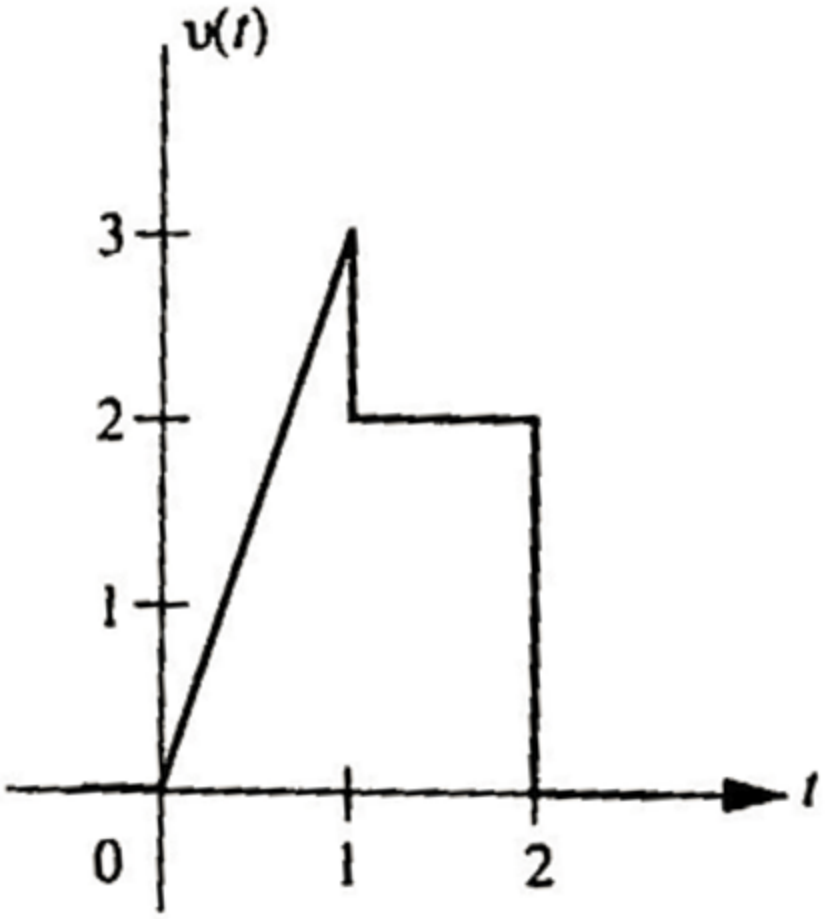


Figure Q5a

5B) Construct t-parameters for the given circuit Figure Q5b (3)

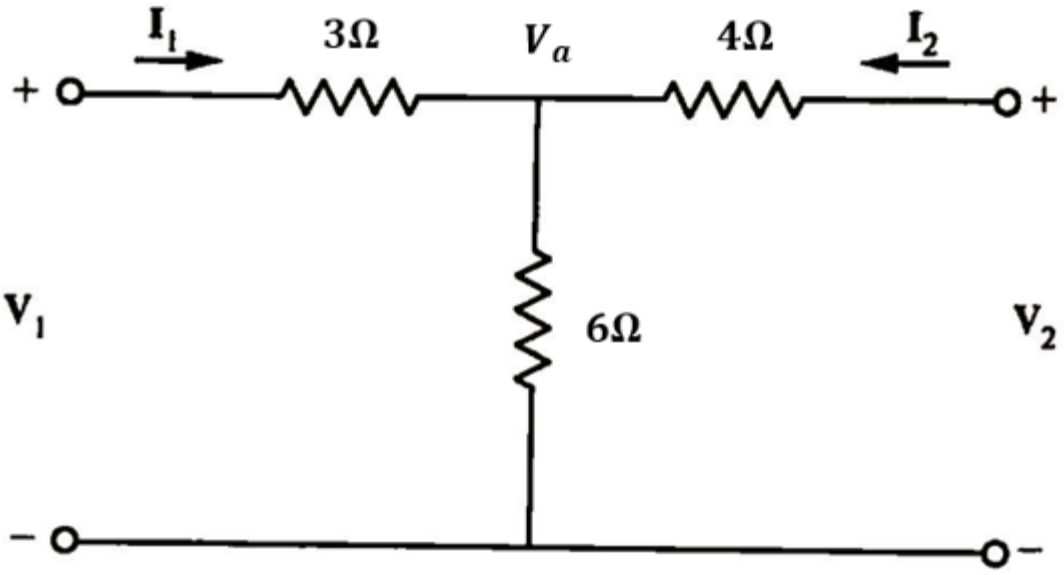


Figure Q5b

5C) Express the voltage pulse shown in Figure Q5c in terms of the unit step function and then evaluate i. $L(v(t))$, ii. $L\left(\frac{dv(t)}{dt}\right)$ (3)

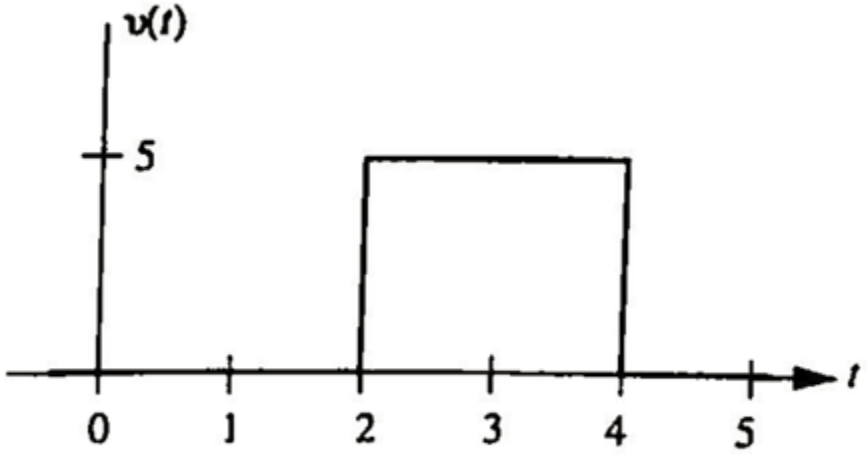


Figure Q5c

-----End-----